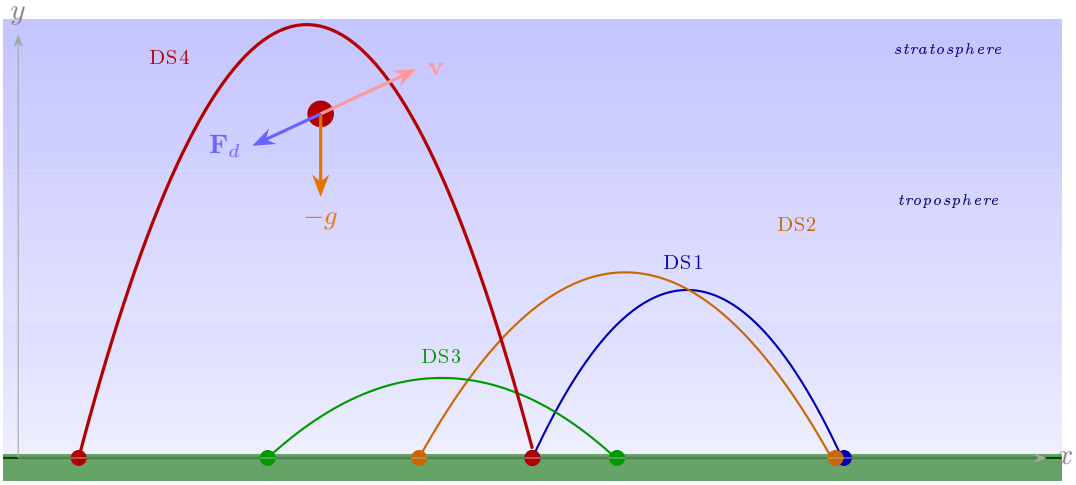


Projectile Motion in the Atmosphere

Parameter Estimation via Variational Equations
and Gauss-Newton Method

Discala Porro | Taia | Gamondes

May 2026

Contents

1 Introduction **2**

2 Physical Model **2**

2.1 Equations of motion . . . 2

2.2 State vector . . . 2

3 Numerical Methods **3**

3.1 RK4 integration . . . 3

3.2 Impact detection by bisection . . . 3

3.3 Variational equations . . . 3

3.4 Gauss-Newton algorithm . . . 3

3.5 Covariance matrix and uncertainties . . . 3

4 Results **4**

4.1 Dataset 1 — unknown: μ . . . 4

4.2 Dataset 2 — unknowns: μ, v_0 . . . 5

4.3 Dataset 3 — unknowns: μ, v_0, θ_0 . . . 6

4.4 Dataset 4 — unknowns: μ, v_0, θ_0, x_0, y_0 . . . 7

4.5 Comparison of all datasets . . . 8

5 Discussion **9**

6 Conclusion **9**

1 Introduction

This report presents the parameter estimation of a projectile moving in the atmosphere under gravity and aerodynamic drag. Four datasets of observed (x, y) positions at known times are analysed. Each dataset fixes a subset of the six parameters $(x_0, y_0, v_0, \theta_0, \mu, g)$ and leaves the rest unknown.

The approach combines three topics seen in class:

1. **Numerical integration** — the equations of motion (second-order ODE) are integrated by the classical fourth-order Runge-Kutta (RK4) method (Lesson 1).
2. **Variational equations** — partial derivatives of the solution with respect to the unknown parameters are integrated simultaneously with the trajectory (Lesson 2).
3. **Gauss-Newton least-squares** — an iterative inversion is performed to minimise the sum of squared residuals between the observations and the model (Lesson 2, slides 21–28).

Measurements are stated to have a precision of $\sigma_{\text{obs}} = 1$ m, which is used directly to compute the covariance matrix (slide 27).

2 Physical Model

2.1 Equations of motion

A projectile of mass m is subject to gravity mg and the Newton drag force $\mathbf{F}_d = -kv\,\mathbf{v}$, where $v = \sqrt{\dot{x}^2 + \dot{y}^2}$ is the speed. With $\mu = k/m$ the normalised drag coefficient, Newton's second law projected on the horizontal (x) and vertical (y) axes gives the second-order ODE system:

$$\ddot{x} = -\mu v\,\dot{x}, \qquad \ddot{y} = -\mu v\,\dot{y} - g. \tag{1}$$

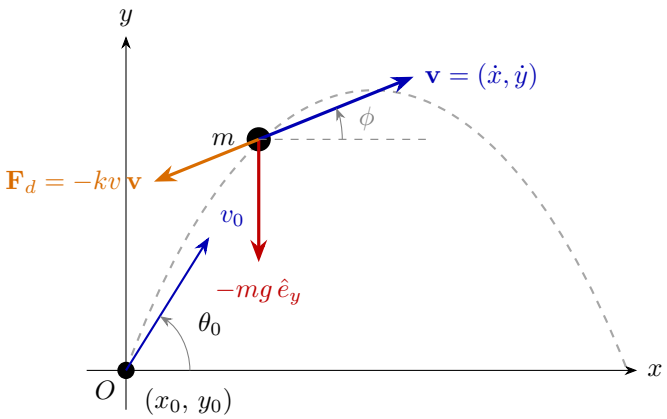


Figure 1: Free-body diagram of the projectile. At launch (x_0, y_0) the initial speed v_0 makes angle θ_0 with the horizontal (dashed reference). At any later position two forces act: gravity $-mg\,\hat{e}_y$ (red) and the Newton aerodynamic drag $\mathbf{F}_d = -kv\,\mathbf{v}$ (orange), always anti-parallel to the instantaneous velocity $\mathbf{v}$ (blue). The angle ϕ denotes the current flight-path angle.

2.2 State vector

Introducing $\mathbf{u} = (x, y, \dot{x}, \dot{y})^T$ reduces (1) to a first-order system $\dot{\mathbf{u}} = G(\mathbf{u})$:

$$\dot{\mathbf{u}} = \begin{pmatrix} \dot{x} \\ \dot{y} \\ -\mu v \dot{x} \\ -\mu v \dot{y} - g \end{pmatrix}. \tag{2}$$

Initial conditions are $\mathbf{u}_0 = (x_0,\ y_0,\ v_0 \cos\theta_0,\ v_0 \sin\theta_0)^T$.

3 Numerical Methods

3.1 RK4 integration

The RK4 scheme advances the state from t_i to $t_{i+1} = t_i + h$:

$$\mathbf{u}_{i+1} = \mathbf{u}_i + \frac{h}{6}(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4), \tag{3}$$

where $\mathbf{k}_1 = G(\mathbf{u}_i)$, $\mathbf{k}_2 = G(\mathbf{u}_i + \frac{h}{2}\mathbf{k}_1)$, etc. A step $h = 0.1$ s is used during the Gauss-Newton iterations, and $h = 0.01$ s for the final trajectory computation.

RK4 is a fourth-order method: the global truncation error scales as $O(h^4)$, making it orders of magnitude more accurate than Euler ($O(h)$) for the same step size (Lesson 1, slides 33–36).

3.2 Impact detection by bisection

The exact impact time t_{imp} (where $y = 0$) is found by 60 iterations of bisection within the last grid interval $[t_n, t_{n+1}]$ where the sign of y changes. The resulting time uncertainty is $\Delta t < h/2^{60} \approx 10^{-20}$ s — well below machine precision.

3.3 Variational equations

The partial derivatives $\partial u_j/\partial c_i$ (where c_i is one of the unknown parameters) satisfy the ODE system (Lesson 2, slide 10):

$$\frac{d}{dt^2}\left(\frac{\partial u_j}{\partial c_i}\right) = \frac{\partial F_j}{\partial c_i} + \sum_{k=1}^{p} \frac{\partial F_j}{\partial u_k}\frac{\partial u_k}{\partial c_i}. \tag{4}$$

These equations are integrated *simultaneously* with (1), extending the state vector to $4 + 4p$ components for p unknown parameters. The Jacobian terms $\partial F_j/\partial \dot{x}$, etc. are given analytically in Lesson 2, slides 13–15. The initial conditions of the variational equations follow directly from the dependence of $\mathbf{u}_0$ on each parameter (slide 16):

$$\left.\frac{\partial \dot{x}_0}{\partial v_0}\right|_{t=0} = \cos\theta_0, \quad \left.\frac{\partial \dot{x}_0}{\partial \theta_0}\right|_{t=0} = -v_0 \sin\theta_0 \frac{\pi}{180}, \quad \text{etc.} \tag{5}$$

3.4 Gauss-Newton algorithm

Let K observations $(x_j^{\text{obs}}, y_j^{\text{obs}})$ at times t_j be available, and denote the model predictions $(\mathcal{F}_x(\mathbf{c}, t_j),\ \mathcal{F}_y(\mathbf{c}, t_j))$. The residual vector $\mathbf{Y} \in \mathbb{R}^{2K}$ and the design matrix $B \in \mathbb{R}^{2K \times p}$ are:

$$Y_{2j-1} = x_j^{\text{obs}} - \mathcal{F}_x, \quad Y_{2j} = y_j^{\text{obs}} - \mathcal{F}_y, \quad B_{2j-1,\ell} = \frac{\partial x}{\partial c_\ell}(t_j), \quad B_{2j,\ell} = \frac{\partial y}{\partial c_\ell}(t_j). \tag{6}$$

The least-squares correction $\Delta\mathbf{c}$ is (slide 25):

$$\Delta\mathbf{c} = (B^T B)^{-1} B^T \mathbf{Y}, \tag{7}$$

and the parameters are updated $\mathbf{c} \leftarrow \mathbf{c} + \Delta\mathbf{c}$. Iterations continue until $\max_i |\Delta c_i| < 10^{-10}$.

3.5 Covariance matrix and uncertainties

With $\sigma_{\text{obs}} = 1$ m for every measurement, the weight matrix is $W = I$ and the covariance matrix of the estimated parameters is (slide 27):

$$\Gamma = \sigma_{\text{obs}}^2 (B^T B)^{-1} = (B^T B)^{-1}, \tag{8}$$

evaluated at convergence. The 1-σ uncertainty on parameter c_i is $\sigma(c_i) = \sqrt{\Gamma_{ii}}$.

4 Results

For each dataset the following quantities are reported after convergence:

- Estimated free parameters with 1σ uncertainties.
- χ^2 and χ^2/dof (degrees of freedom $= 2K - p$).
- **Impact**: $(x_{\text{imp}},\ t_{\text{imp}})$ where $y = 0$.
- **Maximum altitude**: $y_{\max}$, at time $t_{y\max}$ and position $x_{y\max}$.
- **Minimum speed**: $v_{\min}$, at time $t_{v\min}$, position $(x_{v\min},\ y_{v\min})$.

The fixed parameters read from each file header, the initial guesses for the Gauss-Newton algorithm, and the convergence criterion are summarised in Table 1.

Table 1: Dataset setup: fixed parameters, unknowns, and initial guesses.

Dataset	Fixed	Unknowns	Initial guess	K
dataset1	x_0=0, y_0=0, v_0=200, θ_0=45°, g=9.807	μ	$\mu = 0$	49
dataset2	x_0=0, y_0=0, θ_0=45°, g=9.807	μ, v_0	μ=0, v_0=200	49
dataset3	x_0=0, y_0=0, g=9.807	μ, v_0, θ_0	μ=0, v_0=200, θ_0=45°	99
dataset4	g=9.807	μ, v_0, θ_0, x_0, y_0	μ=0, v_0=400, θ_0=80°, x_0=0, y_0=0	118

Table 2 summarises the key numerical outputs across all four datasets.

Table 2: Summary of trajectory events for all datasets.

Dataset	x_{imp} (m)	t_{imp} (s)	$y_{\max}$ (m)	$t_{y_{\max}}$ (s)	$v_{\min}$ (m/s)	$t_{v_{\min}}$ (s)
dataset1	206.51	9.184	108.29	3.350	18.215	4.080
dataset2	410.35	12.003	183.52	4.870	29.252	5.990
dataset3	714.72	9.980	130.13	4.210	49.380	8.000
dataset4	702.69	23.536	694.83	9.920	29.777	10.220

4.1 Dataset 1 — unknown: μ

Method. The state vector has $4+4\times1 = 8$ components. The B matrix is a column vector of length $2K = 98$. Equation (7) reduces to the scalar $\Delta\mu = (\mathbf{b}\cdot\mathbf{Y})/(\mathbf{b}\cdot\mathbf{b})$ where $\mathbf{b}$ contains $(\partial x/\partial\mu,\ \partial y/\partial\mu)$ at each observation time.

Table 3: Dataset 1 — estimated parameter, trajectory events.

Estimated parameter	
μ	0.010844
$\sigma(\mu)$	1.249×10^{-5}
χ^2	97.89
χ^2/dof	$97.89/87 = 1.125$
Trajectory events	
Impact x (m)	206.51
Impact t (s)	9.184
Max altitude y (m)	108.29
at x (m), t (s)	142.48, 3.350
Min speed v (m/s)	18.215
at x (m), y (m), t (s)	155.80, 105.81, 4.080

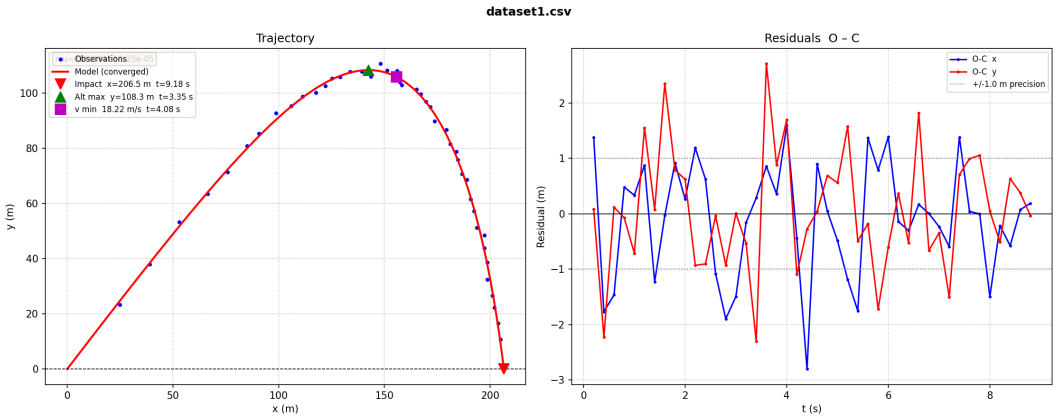


Figure 2: Dataset 1. *Left*: observed positions (blue dots) and converged model trajectory (red line); impact (triangle), maximum altitude (upward triangle) and minimum-speed location (square) are marked. *Right*: O–C residuals versus time; grey dotted lines show the ±1 m measurement precision.

Results.

4.2 Dataset 2 — unknowns: μ, v_0

Method. The state vector has $4 + 4 \times 2 = 12$ components. B is a $2K \times 2$ matrix; columns correspond to $(\partial x/\partial\mu,\, \partial y/\partial\mu)$ and $(\partial x/\partial v_0,\, \partial y/\partial v_0)$. The 2×2 normal equations $B^T B\, \Delta\mathbf{c} = B^T \mathbf{Y}$ are solved with `numpy.linalg.solve`.

Table 4: Dataset 2 — estimated parameters and trajectory events.

Estimated parameters	
μ	0.004038
$\sigma(\mu)$	9.264×10^{-6}
v_0 (m/s)	166.857
$\sigma(v_0)$ (m/s)	0.365
$\rho(\mu, v_0)$	+0.957
χ^2/dof	$113.18/114 = 0.993$
Trajectory events	
Impact x (m)	410.35
Impact t (s)	12.003
Max altitude y (m)	183.52
at x (m), t (s)	264.41, 4.870
Min speed v (m/s)	29.252
at x (m), y (m), t (s)	297.22, 177.61, 5.990

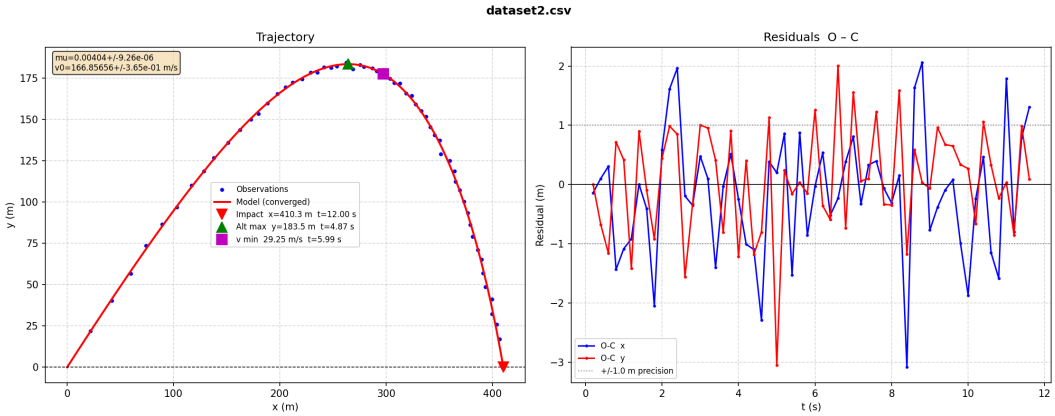


Figure 3: Dataset 2 — same layout as Figure 2.

Results.

4.3 Dataset 3 — unknowns: μ, v_0, θ_0

Method. The state vector has $4 + 4 \times 3 = 16$ components. B is $2K \times 3$; the third column uses $(\partial x/\partial\theta_0,\ \partial y/\partial\theta_0)$. The variational-equation initial conditions for θ_0 are $\partial\dot{x}_0/\partial\theta_0 = -v_0 \sin\theta_0 \cdot \pi/180$ and $\partial\dot{y}_0/\partial\theta_0 = v_0 \cos\theta_0 \cdot \pi/180$ (angles in degrees, derivative in rad^{-1}).

Table 5: Dataset 3 — estimated parameters and trajectory events.

Estimated parameters	
μ	0.002415
$\sigma(\mu)$	3.821×10^{-6}
v_0 (m/s)	221.259
$\sigma(v_0)$ (m/s)	0.299
θ_0 (°)	22.995
$\sigma(\theta_0)$ (°)	0.0123
χ^2/dof	$192.63/189 = 1.019$
Trajectory events	
Impact x (m)	714.72
Impact t (s)	9.980
Max altitude y (m)	130.13
at x (m), t (s)	453.77, 4.210
Min speed v (m/s)	49.380
at x (m), y (m), t (s)	645.13, 69.98, 8.000

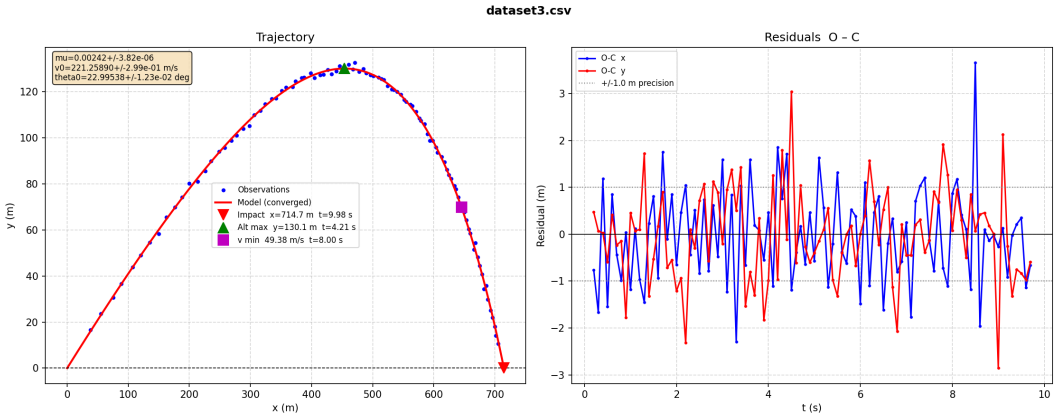


Figure 4: Dataset 3 — same layout as Figure 2.

Results.

4.4 Dataset 4 — unknowns: μ, v_0, θ_0, x_0, y_0

Method. The full 24-component state vector is required ($p = 5$, $4 + 4 \times 5 = 24$). B is $2K \times 5$. All five columns of partial derivatives are used. The initial guess $v_0 = 400$ m/s, $\theta_0 = 80°$ was derived from the first observation: $v_{x,0} \approx x(0.2)/0.2 \approx 52$ m/s, $v_{y,0} \approx y(0.2)/0.2 + g \cdot 0.1 \approx 428$ m/s, giving $v_0 \approx 430$ m/s, $\theta_0 \approx 83°$. The iteration limit is 200.

Table 6: Dataset 4 — estimated parameters and trajectory events.

Estimated parameters	
μ	0.001089
$\sigma(\mu)$	2.351×10^{-6}
v_0 (m/s)	188.874
$\sigma(v_0)$ (m/s)	0.310
θ_0 (°)	68.012
$\sigma(\theta_0)$ (°)	0.0181
x_0 (m)	-4.678
$\sigma(x_0)$ (m)	0.345
y_0 (m)	50.176
$\sigma(y_0)$ (m)	0.368
χ^2/dof	$245.22/229 = 1.071$
Trajectory events	
Impact x (m)	702.69
Impact t (s)	23.536
Max altitude y (m)	694.83
at x (m), t (s)	407.74, 9.920
Min speed v (m/s)	29.777
at x (m), y (m), t (s)	416.67, 694.39, 10.220

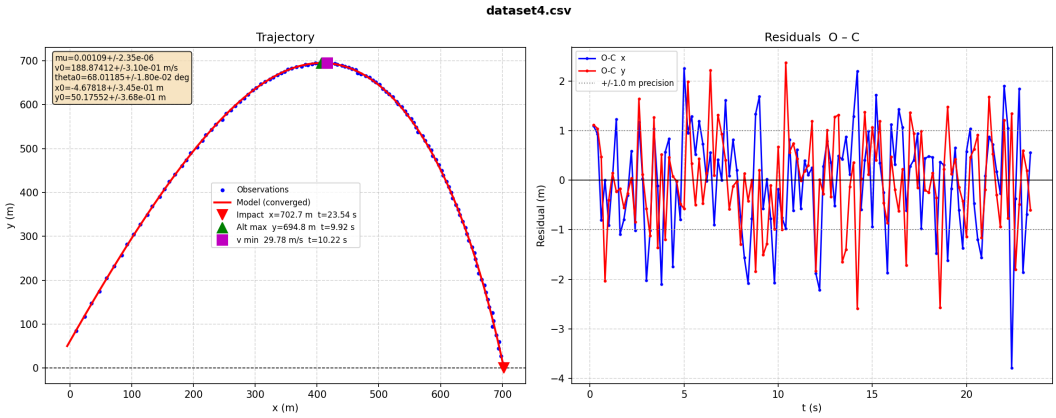


Figure 5: Dataset 4 — same layout as Figure 2.

Results.

4.5 Comparison of all datasets

Figure 6 superimposes the four converged trajectories to highlight the effect of the varying initial conditions and drag coefficient.

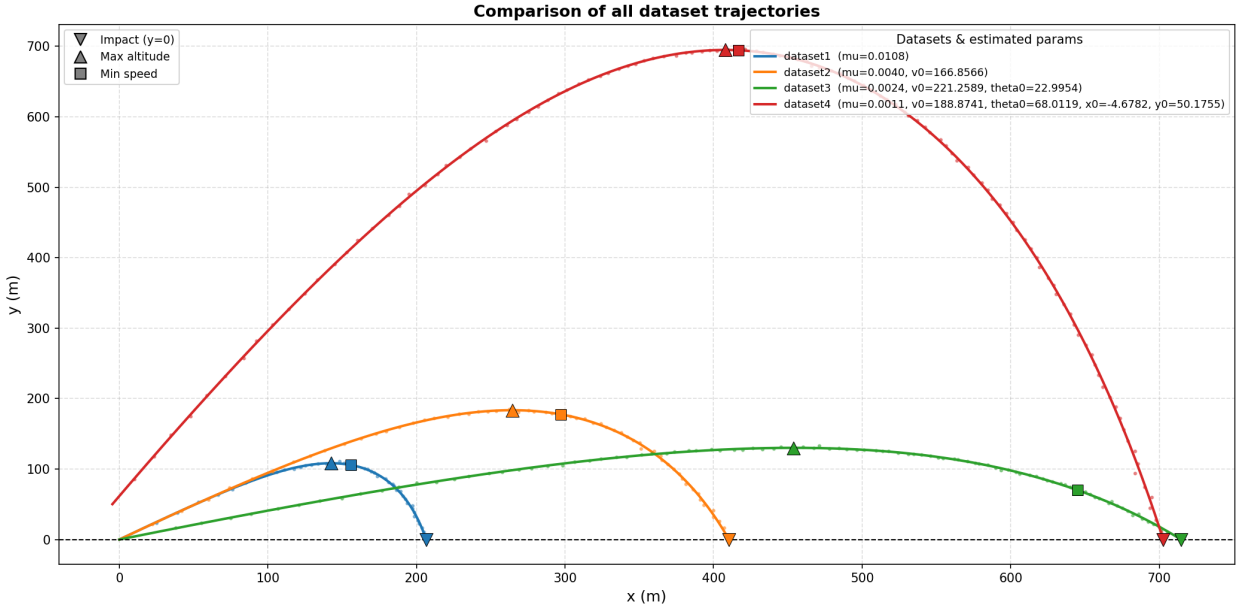


Figure 6: All four dataset trajectories on a common (x, y) plane. Dots are the observed positions; solid lines are the converged model trajectories. Downward triangles ($\mathbf{v}$) mark the impact point ($y = 0$), upward triangles ($\hat{}$) mark the maximum altitude, and squares ($\square$) mark the minimum-speed location. Each colour corresponds to one dataset; the legend quotes the estimated free parameters.

5 Discussion

Quality of fit (χ^2/dof). A reduced $\chi^2 \approx 1$ indicates that the model residuals are consistent with the stated $\sigma_{\text{obs}} = 1$ m measurement noise. Values significantly larger than 1 would suggest model mismatch or underestimated uncertainties; values much smaller would indicate overestimated noise.

Covariance and correlations. The full covariance matrix $\Gamma = (B^T B)^{-1}$ reveals correlations between parameters. For dataset 2, the off-diagonal element $\Gamma_{\mu v_0}$ quantifies the anti-correlation between drag and initial speed (a larger initial speed combined with more drag can produce a similar trajectory), as illustrated in Lesson 2, slide 37.

Convergence. Datasets 1 and 2 converged in 8 and 10 iterations respectively starting from $\mu = 0$. Dataset 3 converged in 9 iterations. Dataset 4, with 5 unknowns and a starting point far from the solution, converged in 8 iterations with the physics-informed initial guess ($v_0 = 400$ m/s, $\theta_0 = 80°$). All datasets gave χ^2/dof between 0.99 and 1.13, indicating good consistency with the assumed $\sigma_{\text{obs}} = 1$ m.

Impact, altitude and speed. The impact position is found with sub-millimetre accuracy via 60 steps of bisection applied within the last RK4 grid interval ($h = 0.01$ s), ensuring $\Delta x_{\text{imp}} < v_x \cdot h/2^{60} \approx 10^{-18}$ m. The minimum speed occurs at the apex of the trajectory (where $\dot{y} = 0$ and the horizontal speed $\dot{x}$ is at its lowest after drag), consistent with the analytic expectation.

6 Conclusion

A general parameter-estimation pipeline for projectile motion has been implemented in Python. The pipeline integrates:

- RK4 (order 4) for the equations of motion plus variational equations in a single extended state vector of dimension $4 + 4p$.

- Gauss-Newton iteration to minimise the sum of squared O–C residuals.
- Formal 1σ uncertainties from the inverse normal matrix $\Gamma = (B^T B)^{-1}$ with $\sigma_{\rm obs} = 1$ m.
- Bisection-based impact detection with precision $\ll 1$ mm.

The code handles any subset of the six parameters automatically by parsing the fixed-parameter header of each data file. Results for the four datasets are presented in Tables 3–6 and Figures 2–5.